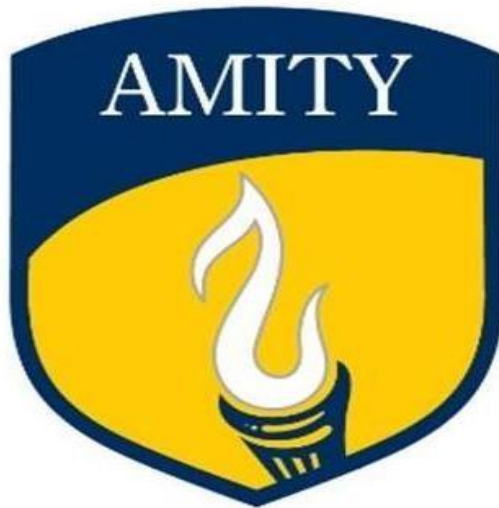


**AMITY UNIVERSITY HARYANA
AMITY SCHOOL OF ENGINEERING AND
TECHNOLOGY**



**APPLIED STATISTICAL ANALYSIS
LAB (AIEU1540)**

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Experiment No	TITLE	Date	Page no	Signature
1	Introduction to R Environment and Basic Commands			
2	Study of Operators in R			
3	Implementation of Decision Making Statements (if-else) in R			
4	Creation and Manipulation of Data Structures (Vector, Matrix, Data Frame, List) in R			
5	Calculation of Measures of Central Tendency (Mean, Median, Mode)			
6	Generation of Frequency and Relative Frequency Tables			
7	Creation of Bar Chart in R			
8	Creation of Pie Chart in R			
9	Representation of Data using Box Plot (Quartiles)			
10	Implementation of Stem and Leaf Plot			
11	Use of Loops in R (for loop / while loop)			
12	Plotting Standard Normal Distribution Curve			
13	Implementation of Binomial Distribution			
14	Implementation of Poisson Distribution			
15	Linear Regression Analysis in R			

Experiment:-1

Aim: To familiarize with the R programming environment and execute basic commands.

Theory: R is a programming language and software environment for statistical computing and graphics. It provides a wide variety of statistical and graphical techniques, and is highly extensible

Code:

```
x <- 10
y <- 5
sum_val <- x + y
print(sum_val)
getwd()
ls()
```

Output:

```
[1] 15
[1] "C:/Users/abhis/OneDrive/Documents"
[1] "sum_val" "x"      "y"
```

Experiment:-2

Aim: To implement and understand various arithmetic, relational, and logical operators in R.

Theory: Operators are used to perform operations on variables and values. R includes arithmetic operators (+, -, *, /), relational operators (>, <, ==) , and logical operators (&, |, !).

Code:

```
a <- 20  
b <- 10  
  
print(a * b)  
print(a > b)  
  
vec1 <- c(TRUE, FALSE)  
vec2 <- c(TRUE, TRUE)  
print(vec1 & vec2)
```

Output:

```
[1] 200  
  
[1] TRUE  
  
[1] TRUE FALSE
```

Experiment:-3

Aim: To control the flow of execution using conditional statements.

Theory: Decision-making structures require the evaluation of one or more conditions. The if-else statement executes a block of code if a condition is true, and another if it is false.

Code:

```
num <- 42  
if (num %% 2 == 0) {  
  print("The number is even")  
}  
else { print("The number is odd") }
```

Output:

```
[1] "The number is even"
```

Experiment:-4

Aim: To create and manipulate Vectors, Matrices, Data Frames, and Lists.

Theory: R's primary data structures include vectors (1D arrays), matrices (2D arrays), data frames (tablelike structures), and lists (collections of different object types).

Code:

```
vec <- c(1, 2, 3)
mat <- matrix(1:6, nrow=2, ncol=3)
df <- data.frame(ID = 1:2, Name = c("A", "B"))
my_list <- list(vec, mat, df)
print(my_list)
```

Output:

```
[[1]]
[1] 1 2 3

[[2]]
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

[[3]]
  ID Name
1  1   A
2  2   B
```

Experiment:-5

Aim: To calculate the Mean, Median, and Mode of a dataset.

Theory: Central tendency represents the center of a distribution. Mean is the average, median is the middle value, and mode is the most frequent value.

Code:

```
data_set <- c(12, 15, 12, 18, 20, 12, 25)
mean_val <- mean(data_set)
median_val <- median(data_set)
get_mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
mode_val <- get_mode(data_set)
print(paste("Mean:", mean_val))
print(paste("Median:", median_val))
print(paste("Mode:", mode_val))
```

Output:

```
[1] "Mean: 16.2857142857143"
[1] "Median: 15"
[1] "Mode: 12"
```

Experiment:-6

Aim: To organize raw data into frequency and relative frequency distributions.

Theory: Frequency is the count of occurrences. Relative frequency is the frequency divided by the total number of observations.

Code:

```
data <- c("A", "B", "A", "C", "B", "A", "A")
freq_table <- table(data)
rel_freq_table <- prop.table(freq_table)
print(freq_table)
print(rel_freq_table)
```

Output:

```
data
A B C
4 2 1

data
      A      B      C
0.5714286 0.2857143 0.1428571
```


Experiment:-7

Aim: To visualize categorical data using a bar chart.

Theory: A bar chart uses rectangular bars to represent the frequency or value of different categories.

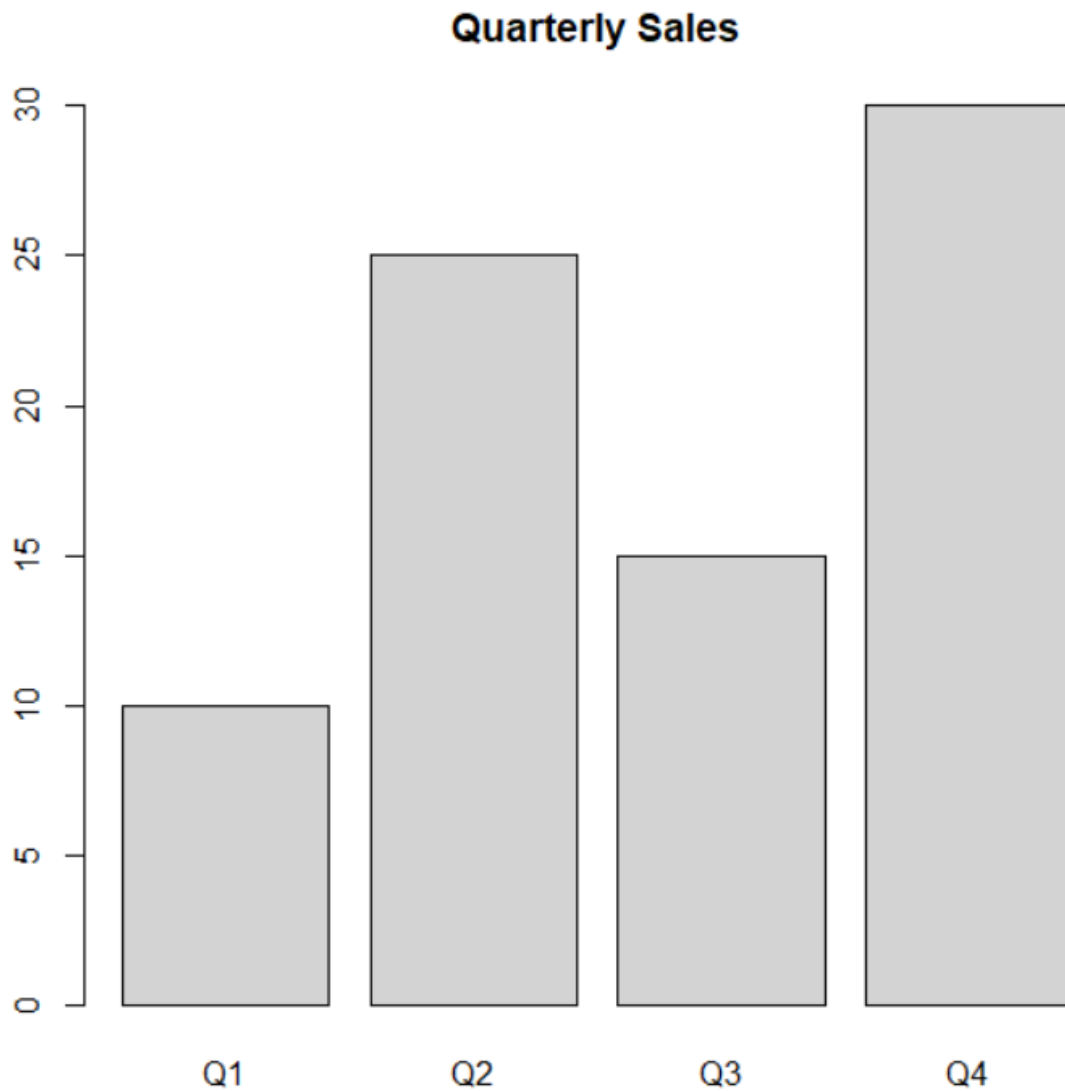
Code:

```
values <- c(10, 25, 15, 30)
```

```
labels <- c("Q1", "Q2", "Q3", "Q4")
```

```
barplot(values, names.arg = labels, col = "lightgray", main = "Quarterly Sales")
```

Output:



Experiment:-8

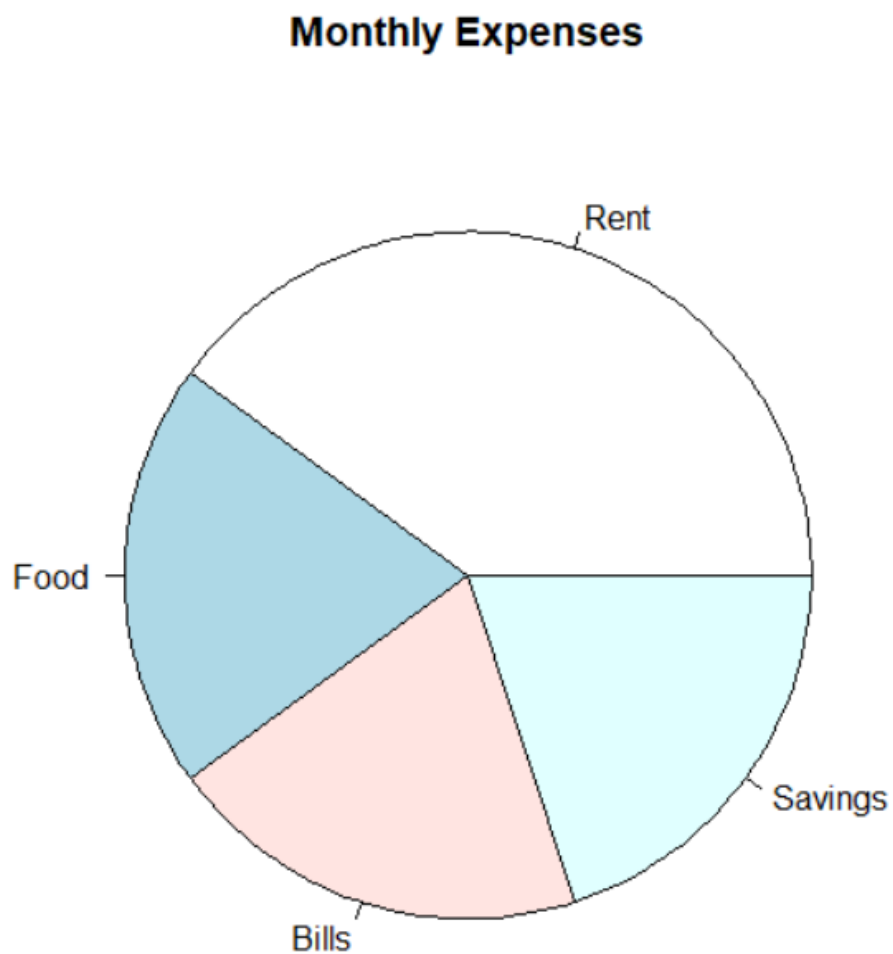
Aim: To represent data proportions using a pie chart.

Theory: A pie chart is a circular plot where each 'slice' represents a proportion of the total.

Code:

```
slices <- c(40, 20, 20, 20)
labels <- c("Rent", "Food", "Bills", "Savings")
pie(slices, labels = labels, main = "Monthly Expenses")
```

Output:



Experiment:-9

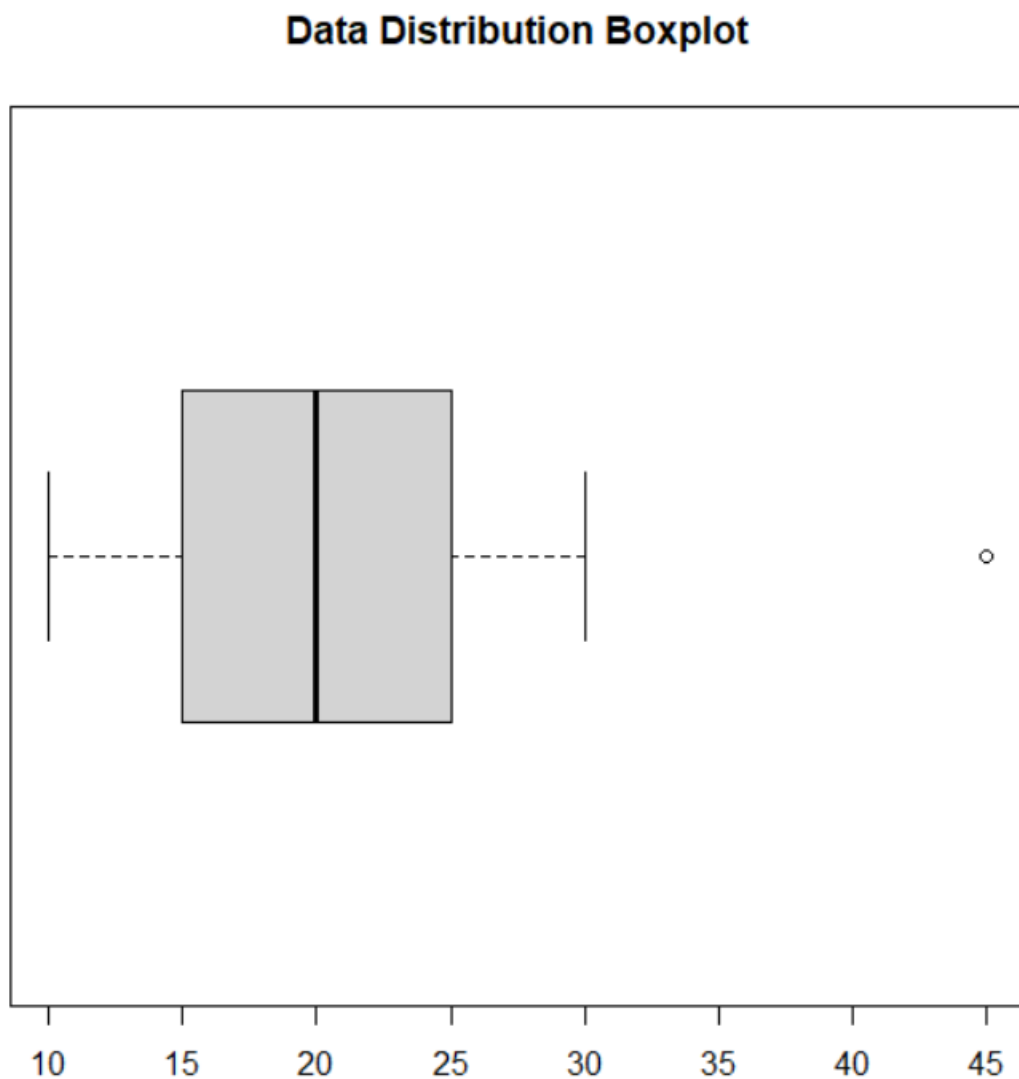
Aim: To visualize the five-number summary and quartiles using a box plot.

Theory: A box plot summarizes data using the minimum, first quartile, median, third quartile, and maximum.

Code:

```
stats_data <- c(10, 12, 15, 18, 20, 22, 25, 30, 45)
boxplot(stats_data, main = "Data Distribution Boxplot", horizontal = TRUE)
```

Output:



Experiment:-10

Aim: To create a stem-and-leaf plot for a numeric dataset.

Theory: A stem-and-leaf plot is a textual representation of quantitative data that shows individual data points while revealing the distribution shape.

Code:

```
data_points <- c(12, 15, 18, 22, 25, 26, 31, 33, 33, 45)
stem(data_points)
```

Output:

The decimal point is 1 digit(s) to the right of the |

```
The decimal point is 1 digit(s) to the right of the |
```

```
1 | 258
2 | 256
3 | 133
4 | 5
```

Experiment:-11

Aim: To implement iterative logic using for and while loops.

Theory: Loops allow for repeated execution of code blocks. For loops iterate over a sequence, while loops continue as long as a condition is met.

Code:

```
for (i in 1:3) {  
  print(paste("Iteration", i))  
}  
  
count <- 1  
while (count <= 2) {  
  print(paste("Count:", count))  
  count <- count + 1  
}
```

Output:

```
[1] "Iteration 1"  
[1] "Iteration 2"  
[1] "Iteration 3"  
  
[1] "Count: 1"  
[1] "Count: 2"  
> |
```

Experiment:-12

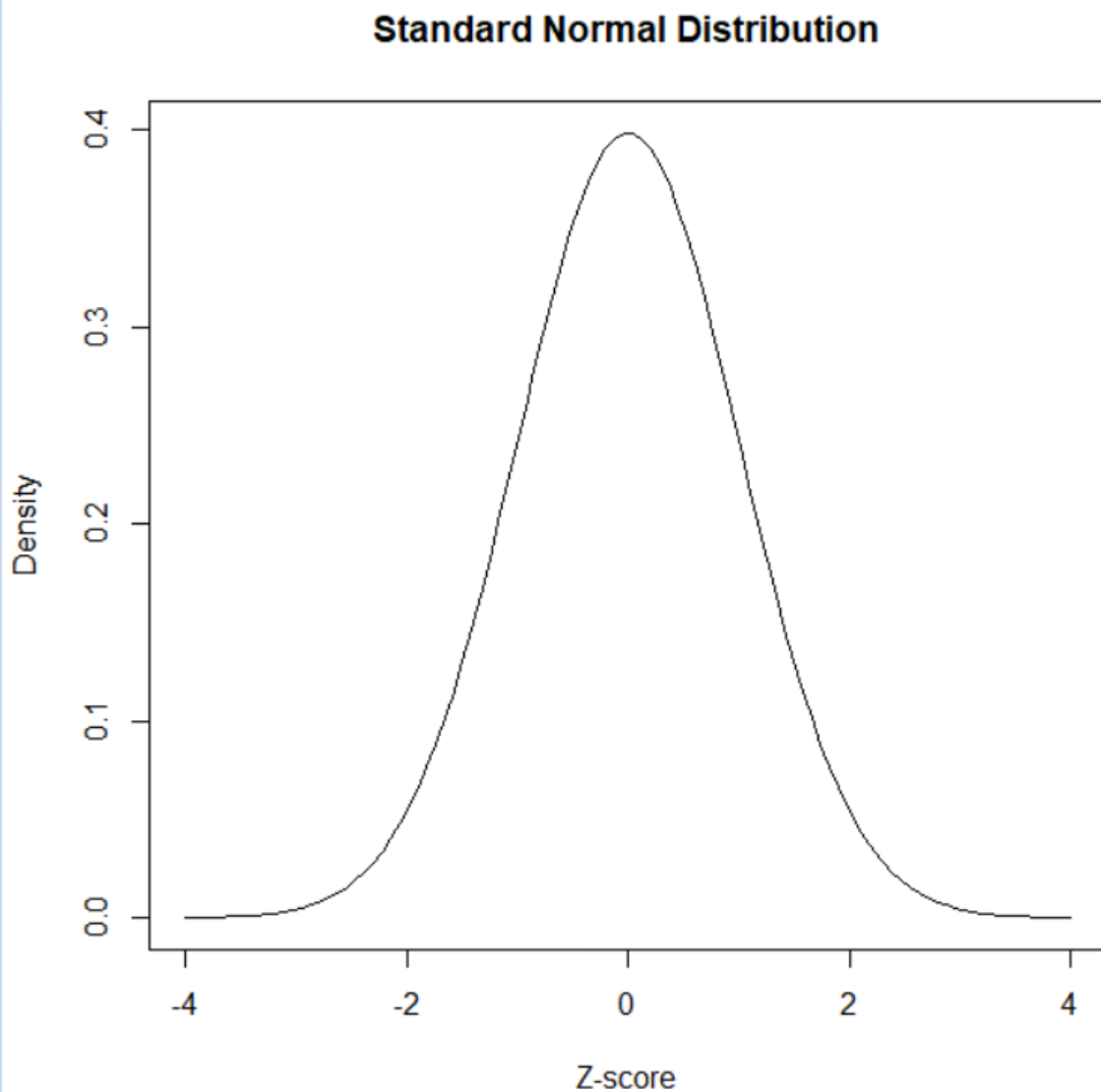
Aim: To generate and plot a standard normal distribution curve.

Theory: The standard normal distribution is a bell-shaped curve with mean = 0 and standard deviation = 1.

Code:

```
x <- seq(-4, 4, length=100)
y <- dnorm(x, mean=0, sd=1)
plot(x, y, type="l", main="Standard Normal Distribution", xlab="Z-score", ylab="Density")
```

Output:



Experiment:-13

Aim: To calculate probabilities using the Binomial distribution.

Theory: The binomial distribution models the number of successes in n independent trials with a constant probability of success p .

Code:

```
prob_3 <- dbinom(3, size = 10, prob = 0.5)
print(prob_3)
```

Output:

```
[1] 0.1171875
> |
```

Experiment:-14

Aim: To calculate probabilities using the Poisson distribution.

Theory: The Poisson distribution models the number of events in a fixed interval given a constant average rate (λ).

Code:

```
prob_pois <- dpois(2, lambda = 3)
print(prob_pois)
```

Output:

```
[1] 0.2240418
> |
```


Experiment:-15

Aim: To perform linear regression and determine the relationship between variables.

Theory: Linear regression fits a straight line to data points to model the relationship between an independent variable and a dependent variable.

Code:

```
height <- c(151, 174, 138, 186, 128)
```

```
weight <- c(63, 81, 56, 91, 47)
```

```
model <- lm(weight ~ height)
```

```
print(summary(model))
```

Output:

```
Call:
lm(formula = weight ~ height)

Residuals:
    1      2      3      4      5 
-1.3273 -0.4347  1.3421  0.6397 -0.2199 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -47.98667     3.82989  -12.53   0.0011 **
height        0.74380     0.02441   30.47 7.76e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.185 on 3 degrees of freedom
Multiple R-squared:  0.9968,    Adjusted R-squared:  0.9957 
F-statistic: 928.6 on 1 and 3 DF,  p-value: 7.763e-05
```